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Topic: Industrial energy use and emission trends in Ontario's broader public sector (2011 through 2023)

Objective: To determine whether emissions from industrial public operation sector buildings are more volatile over time compared to other public operation sectors, and higher in volume. Analysis will be done to determine key differences over time between operation sectors.

Data Source and Details:

<https://data.ontario.ca/dataset/energy-use-and-greenhouse-gas-emissions-for-the-broader-public-sector> (2011 - 2023)

Pre-cleaned records: 212,271

Pre-cleaned data size: 21.1 MB

Post-cleaned records: 178,555

Post-cleaned data size: 23.2 MB

Attributes/Features:

- Year
- Sector
- Subsector
- Organization
- Operation
- Operation Type
- City
- Postal Code
- Property General Floor Area (GFA) m²
- Weekly Average Hours
- Source EUA (GJ/m²)
- GHG Emissions (1000 kg CO₂e)
- GHG Emissions Intensity (kg CO₂e/m²)

Engineered Attributes/Features:

- Normalized GHG Emissions Intensity (kg CO₂e/m²hr)
- GHG Emissions Intensity Recalculated (kgCO₂e/m²)
- Operation Sector

The dataset consists of the Government of Ontario's 2011 through 2023 Energy Use and Greenhouse Gas Emissions (GHG) for the Broader Public Sector (BPS) datasets. This dataset is of particular interest to the team as emissions tracking over time and analysis of industrial processes allows some of us to pursue work in our fields of study (environment and geography), and for everyone to apply data science and analysis techniques learned in this course.

Methods:

The concatenated dataset was created through web scraping and Pandas. Data, including large outliers, was cleaned using imputed values wherever possible. Leftover null data that

could not be imputed was subsequently removed. Engineered attributes were created such that proper comparisons among Operation Sectors could be performed. Analysis commenced using matplotlib, seaborn, scipy, statsmodels, and patsy in which 'GHG Emissions Intensity Recalculated' and 'Normalized GHG Emissions Intensity' were studied. This analysis included key statistical metrics, such as skewness and normality, data correlations, percent change graphing and statistics, and linear regression modeling of key predictors, such as Operation Sector and Year.

Analysis:

Challenge 1: One of the largest challenges presented to the team was the data cleaning. Due to the volume of data and many missing or erroneous data values, imputations were made wherever possible, however significant variation in the dataset remained. The challenge occurred where Property GFA (gross floor area) values were sometimes provided for industrial processes as 0.00001 m² or Average Weekly Hours were provided as 0.01 hours. This information could not easily be fixed via imputation because many similar entries in the same Operation Sectors, such as industrial, contained these low values, and the team did not want to impute values given this was a core metric used to normalize the data. To address this issue where imputation was not possible, data below thresholds of 1 m² and greater than 0.0 hours were used to retain as much data as possible and minimize decimal values greatly skewing the normalized GHG emissions intensity data. It is believed that these erroneous values occurred due to user error while inputting information upon the Government of Ontario's request.

Challenge 2: A secondary challenge to the team was figuring out the driving force of the emissions results. While the team was able to answer the question to the null hypothesis of industrial emissions being more volatile than that of other Operation Sectors, identifying the driving factors that caused these emissions was not possible given the dataset attributes. There were no correlations with the numerical inputs (property size and operating hours) and most of the categorical variables (sector, subsector, operation, city) all showed very strong relationships to the outputs. However, there was no obvious trend within those variables and more advanced statistical modeling might be needed to interpret the results better. Further attempts to find driving factors were made by linear regression models using a variety of attributes, but to no avail. Merging of other related datasets and a deeper understanding of the inputs would be required to determine the driving factors.

Conclusions:

The team's analysis concludes that there have been modest declines in greenhouse gas emissions in Ontario's broader public sector from 2011 to 2023. The analysis also points to both greater emissions output and greater volatility of emissions in the industrial operating sector as compared to other public operating sectors. However, exploration and analysis of the data brought to light serious limitations in interpreting the data in any sort of conclusive fashion due to the range of extreme values likely due to errors in data entry made by building managers completing the survey. It is recommended that the Ontario government engage in more activities to accelerate decarbonization of the public sector. It is also recommended that they invest more in automation of the data entry process in order to be able to draw more definitive conclusions around this vital area of public interest.